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CODE:

```
# Implementation of FIND-S Algorithm

# Training data:
# Each sample contains attributes followed by the target concept
# "Yes" = Positive example
# "No"  = Negative example

training_data = [
    ["Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"],
    ["Sunny", "Warm", "High", "Strong", "Warm", "Same", "Yes"],
    ["Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"],
    ["Sunny", "Warm", "High", "Strong", "Cool", "Change", "Yes"]
]

# Number of attributes (excluding target)
num_attributes = len(training_data[0]) - 1

# Step 1: Initialize the most specific hypothesis
hypothesis = ["0"] * num_attributes

print("Initial Hypothesis:")
print(hypothesis)

# Step 2: Process each training example
for example in training_data:

    attributes = example[:-1]
    target = example[-1]

    # FIND-S considers only positive examples
    if target == "Yes":

        for i in range(num_attributes):

            # If hypothesis is most specific, replace with attribute value
            if hypothesis[i] == "0":
                hypothesis[i] = attributes[i]

            # If values are different, generalize using '?'
            elif hypothesis[i] != attributes[i]:
                hypothesis[i] = "?"

    print("\nTraining Example:", example)
    print("Current Hypothesis:", hypothesis)

# Final hypothesis
print("\nFinal Most Specific Hypothesis:")
print(hypothesis)
```

OUTPUT :

```
Initial Hypothesis:
['0', '0', '0', '0', '0', '0']

Training Example: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes']
Current Hypothesis: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']

Training Example: ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes']
Current Hypothesis: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Training Example: ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No']
Current Hypothesis: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']

Training Example: ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
Current Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

Final Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```